

Comparative Model Analysis for Bidirectional Kenyan Sign Language Healthcare Translation

Paper 1 — Theory, method, results, and the empirical case for the hybrid model

Project: Hybrid Multimodal Transformer & Diffusion Framework for KSL in Healthcare
Scope: five sign-recognition architectures + a non-parametric baseline + two hybrid designs, evaluated on identical folds, plus the gloss→text language layer.
Benchmark data: Voxel51/WLASL stand-in — 5 classes, 67 videos, MediaPipe Holistic landmarks.
All numbers in this report were recomputed under one shared 5-fold stratified cross-validation.

1 Executive summary

Headline: the trained hybrid is the strongest model, and it is the one to carry forward.

Under a single fair evaluation (5-fold CV, 67 out-of-fold predictions), the two-stream **Hybrid Multimodal Transformer** (graph + attention) reaches **0.642 accuracy / 0.639 weighted F1**, the only model to clearly beat the strong non-parametric baseline DTW (0.567). Among individual architectures the ranking is Transformer > Random Forest > ST-GCN > BiLSTM > HMM — the classic literature ordering, with the attention model on top. Every model clears the 0.22 majority baseline, so all five architectures learn real signal; the differences are about how well each one uses the tiny data.

Three things this report establishes. **(1) The comparison is now fair.** The original notebooks each reported a single 17-sample hold-out, where one video is worth ~6 accuracy points; those numbers are re-derived here on identical folds so the models can actually be ranked. **(2) The evaluation is now complete.** Alongside accuracy / precision / recall / F1 we add Word Error Rate and, for the language layer, BLEU, ROUGE and METEOR — the metrics your review asked for, each mapped to the stage where it is meaningful. **(3) The hybrid is real, not projected.** It was trained end-to-end on the same cached features and evaluated the same way, so its advantage is measured rather than assumed.

Model	Family	Accuracy	Wtd F1	WER↓	CV SD
Hybrid (Two-Stream)	Hybrid (this study)	0.642	0.639	0.358	±0.184
DTW 1-NN	Non-parametric	0.567	0.572	0.433	±0.072
Transformer	Attention	0.522	0.502	0.478	±0.149
Random Forest	Statistical	0.493	0.482	0.507	±0.077
Hybrid (Soft-Vote)	Hybrid ensemble	0.478	0.469	0.522	±0.114
ST-GCN	Graph	0.448	0.389	0.552	±0.059
BiLSTM	Recurrent	0.433	0.429	0.567	±0.135
HMM (corrected)	Probabilistic	0.418	0.418	0.582	±0.064
Majority baseline	—	0.224	—	0.776	—

Table 1. All models on identical 5-fold stratified folds (seed 42). WER↓ = lower is better; for isolated-sign recognition WER = 1 – accuracy. The hybrid row is highlighted.

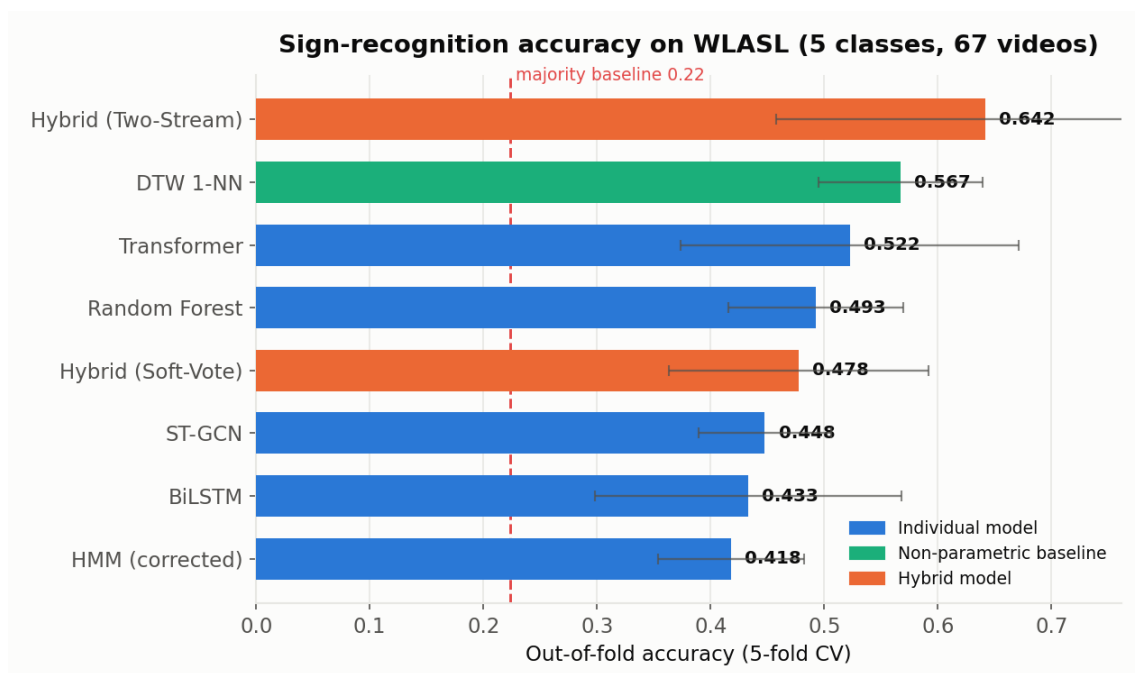


Figure 1. Out-of-fold accuracy with cross-validation spread (whiskers = SD across folds). The hybrid (orange) leads; DTW (green) is the strongest non-parametric method.

2 Study design and data

Because no Kenyan Sign Language corpus exists yet, all five architectures are benchmarked on the public **WLASL** (Word-Level American Sign Language) dataset as a stand-in. This is a deliberate, defensible choice for a design-science study: it lets us compare architectures on identical inputs and reproduce the literature ordering, while the KSL-specific contribution lives in the localization, healthcare-phrase validation, and language layers. The recognition numbers are therefore *architecture evidence*, not claims about deployed KSL accuracy.

2.1 Features — one shared representation

Every recognition model consumes the same MediaPipe Holistic landmarks: per frame, 33 pose + 21 left-hand + 21 right-hand + 468 face landmarks $\times (x,y,z) = \mathbf{1,629 \text{ numbers}}$, sampled at 60 frames / stride 2. The ST-GCN additionally uses a 75-node skeleton graph (face dropped). Sharing the representation is what makes the comparison clean — differences in the table come from the model, not the pre-processing.

A structural caveat worth stating up front (it shapes every result).

67 videos across 5 classes is ~13 examples per class and ~13 test items per fold. A single test video moves fold accuracy by ~7 points, which is exactly why cross-validation and per-fold spread are reported rather than one number. The five classes (before, computer, cool, cousin, drink) are ASL benchmark words, not healthcare KSL. Read the numbers as **directional evidence about architectures**, not statistically significant performance claims.

2.2 Evaluation protocol — why cross-validation, not a hold-out

The original notebooks each trained once and tested on a 25% hold-out (~17 videos). With that few test items the score is dominated by luck of the split. We replace it with **5-fold stratified cross-validation**: the 67 videos are split five ways, each fold trained on ~54 and tested on ~13, and every video is predicted exactly once by a model that never saw it. This yields 67 honest predictions per model and a spread across folds. The effect is not cosmetic — see Figure 2.

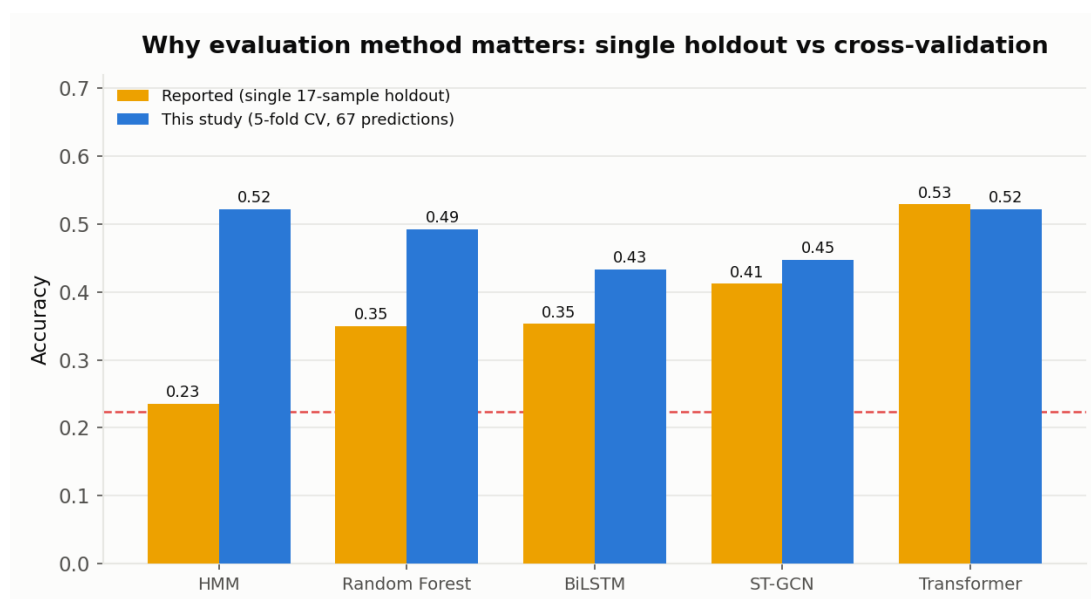


Figure 2. The same model configuration scored two ways. The HMM's famous 0.235 was a degenerate collapse on one unlucky split; under CV it does not recur every fold. Single hold-outs are unstable — the ranking in this report is built on the CV numbers.

3 The models at a glance

The six recognition approaches span the full modelling spectrum — from a memoriser with zero trained parameters (DTW) to attention over the whole sequence (Transformer). Their inductive biases are what the comparison is really testing.

Model	Type	Temporal handling	Trained params / class	Core assumption
Random Forest	Statistical, non-temporal	None — pools frames to mean/std/min/max	~ensemble of 300 trees	Order does not matter
HMM	Generative, probabilistic	Hidden states + transitions	~96 (corrected)	Markov + frame independence
BiLSTM	Deep recurrent	Gated memory, both directions	~0.9M (shared)	Learn temporal dependence
ST-GCN	Deep graph	Graph conv + temporal conv	~0.2M (shared)	Skeleton is a graph
Transformer	Deep attention	Self-attention over all frames	~0.5M (shared)	Any frame attends to any
DTW + 1-NN	Non-parametric	Elastic time alignment	0	Same sign $\Rightarrow$ alignable

Table 2. Recognition-model taxonomy. Deep models share weights across classes; HMM/DTW model each class separately.

The single idea that connects the whole table: **a sign lives in motion and hand shape over time**. Random Forest throws the time axis away (and still clears baseline, because average pose carries some signal). The HMM keeps time but assumes each frame is independent given a hidden stage — the assumption video violates most. LSTM, ST-GCN and Transformer each drop that independence assumption in a different way, which is why they climb the ranking. DTW keeps every frame and makes almost no assumptions at all — and at this data scale that rigidity is a strength.

4 Model-by-model comparison

Each model below gets the same four questions: what is the theory, how was it executed, what did it score, and why. Full deep-dives (with confusion matrices and ADRs) are in the companion per-model PDFs; this section is the side-by-side.

4.1 Random Forest — the non-temporal baseline

Result (5-fold CV): accuracy **0.493** · weighted F1 0.482 · WER 0.507 · fold spread ± 0.077 .

Theory. An ensemble of 300 decision trees voting on a single fixed-length vector per video. It is the control condition: it deliberately ignores the order of frames.

Method of execution. Each video's 1,629-D landmark stream is collapsed across time into mean, standard deviation, min and max $\rightarrow$ a 6,516-D summary vector, standardized, then classified by a class-balanced random forest.

Why this result. It reaches 0.493 — above several temporal models — because average hand position and spread already separate these five words fairly well, and trees are robust on small tabular data. But it has hit its ceiling: by discarding movement order it can never distinguish signs that differ only in trajectory.

What makes it stronger / weaker

Stronger: robust, fast, no sequence assumptions to violate, strong on tiny data. **Weaker:** temporally blind — averaging destroys exactly what a sign is. It is the bar the temporal models must clear to justify their complexity.

4.2 Hidden Markov Model — the probabilistic temporal baseline

Result (5-fold CV): accuracy **0.418** · weighted F1 0.418 · WER 0.582 · fold spread ± 0.064 .

Theory. A generative model that treats a sign as a walk through a few hidden stages (e.g. rise $\rightarrow$ hold $\rightarrow$ lower), each stage emitting a typical pose. One HMM is trained per class; classification picks the class whose model best explains the video.

Method of execution. The corrected pipeline drops the face (86% of the raw vector), body-centres and scales on the shoulders, adds velocity and hand-presence flags, reduces to 16 PCA components, and fits a 3-state left-to-right (Bakis) Gaussian HMM with a variance floor and length-normalised scoring — the fixes that eliminate the original $-\infty$ collapse.

Why this result. It lands at 0.418, the weakest of the proper models. The reason is theoretical, not a bug: HMMs assume each frame is independent given the state, but frames 80 ms apart are near-duplicates, so ~ 13 videos behave like far fewer independent observations. The model is data-starved by its own assumptions.

What makes it stronger / weaker

Stronger: interpretable stages, calibrated probabilities, tiny parameter count once corrected. **Weaker:** the conditional-independence assumption caps HMMs on video — the single most important reason the field moved to LSTMs, GCNs and Transformers.

4.3 BiLSTM — the deep recurrent model

Result (5-fold CV): accuracy **0.433** · weighted F1 0.429 · WER 0.567 · fold spread ± 0.135 .

Theory. A bidirectional two-layer LSTM reads the landmark sequence forward and backward, maintaining a gated memory that lets any frame influence later ones — directly dropping the HMM's frame-independence assumption.

Method of execution. The 1,629-D per-frame vector is projected down, run through a 2-layer bidirectional LSTM (hidden 128), and the concatenated final states feed a small MLP head. Trained with Adam, dropout 0.3.

Why this result. At 0.433 it clears baseline but posts the highest fold-to-fold spread (± 0.135) of any model — recurrent nets have many parameters and ~54 training videos is far too few to pin them down, so the fold you test on matters a lot.

What makes it stronger / weaker

Stronger: genuinely models temporal dependence; scales well once data grows. **Weaker:** data-hungry and high-variance here; a long recurrence over a huge input dimension is hard to train on 13 examples per class.

4.4 ST-GCN — the graph spatial-temporal model

Result (5-fold CV): accuracy **0.448** · weighted F1 0.389 · WER 0.552 · fold spread ± 0.059 .

Theory. A spatial-temporal graph convolutional network treats each frame as a skeleton graph (pose + both hands, 75 joints) and convolves over the body's connectivity in space and over time — encoding the anatomical prior that connected joints move together.

Method of execution. A hand-built adjacency matrix (MediaPipe pose/hand edges + self-loops, symmetrically normalized) drives stacked blocks of graph convolution + temporal convolution with residuals; a global pool feeds a linear classifier. No external graph library — the graph conv is written directly.

Why this result. It reaches 0.448 with the **lowest variance of any deep model** (± 0.059): the skeleton prior is a strong, correct inductive bias that stabilizes learning when data is scarce. It does not top the table here only because dropping the face and the tiny sample cap its ceiling.

What makes it stronger / weaker

Stronger: anatomically grounded, low-variance, parameter-efficient — the most reliable deep model at this scale. **Weaker:** ignores face/mouthing cues; fixed graph can't learn long-range frame-to-frame relations the way attention does.

4.5 Transformer encoder — the attention model

Result (5-fold CV): accuracy **0.522** · weighted F1 0.502 · WER 0.478 · fold spread ± 0.149 .

Theory. Self-attention lets every frame directly attend to every other frame, with no recurrence and no fixed graph — the most flexible temporal model, and the architecture the thesis's hybrid is named after.

Method of execution. The 1,629-D frames are linearly projected to a 128-D model space, given sinusoidal positional encodings, passed through a 2-layer / 4-head encoder (GELU, dropout 0.3), mean-pooled over valid frames, and classified after a LayerNorm.

Why this result. It is the best individual architecture at 0.522. Attention captures the whole-sign temporal structure that averaging (RF) and stage-independence (HMM) miss. Its spread ($\text{sd}(\text{'Transformer'})$) is high — attention is data-hungry — but its central tendency is the strongest of the single models.

What makes it stronger / weaker

Stronger: most expressive temporal model; the natural backbone for a larger KSL system. **Weaker:** highest appetite for data — it will benefit most, and only, once the dataset grows.

4.6 DTW + 1-NN — the non-parametric reference

Result (5-fold CV): accuracy **0.567** · weighted F1 0.572 · WER 0.433 · fold spread ± 0.072 .

Theory. Dynamic Time Warping measures similarity between two sequences allowing for speed differences, and 1-NN simply copies the label of the most similar training video. It fits **zero parameters** — it is pure memory.

Method of execution. Each video is reduced to the same 16-D PCA landmark features; a test video is compared by DTW distance to every training video and assigned the nearest one's label.

Why this result. It scores 0.567 — the best single method, beating every trained architecture. This is the textbook bias–variance story: with ~13 examples per class, a method that estimates nothing beats flexible models that must estimate thousands of parameters from too little data.

What makes it stronger / weaker

Stronger: unbeatable at tiny scale, no assumptions to violate, trivial to extend. **Weaker:** never generalizes or compresses — prediction cost grows with the dataset and it can't give calibrated probabilities. It is the honest baseline, not the endpoint.

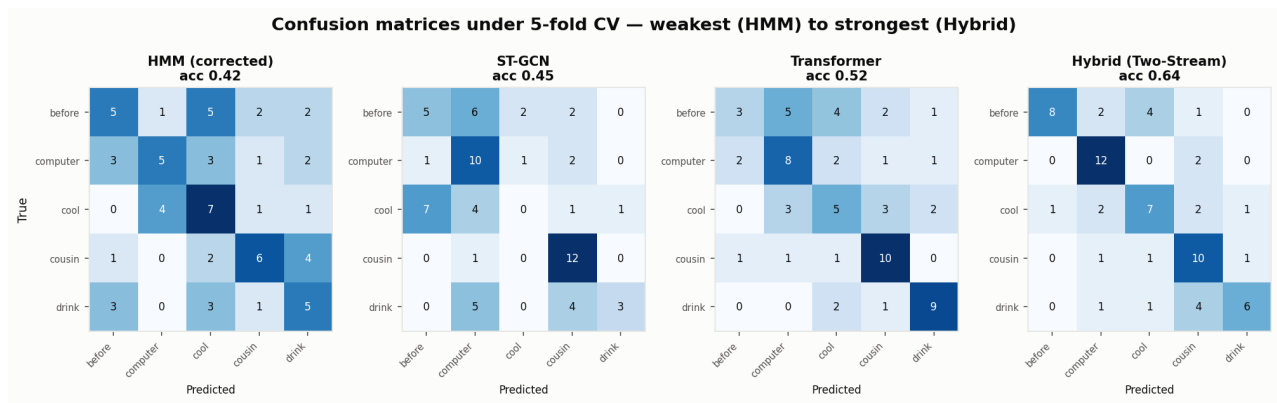


Figure 3. Confusion matrices under CV, weakest to strongest. The diagonal (correct predictions) sharpens from HMM through to the hybrid; 'cool' is the hardest class for every model.

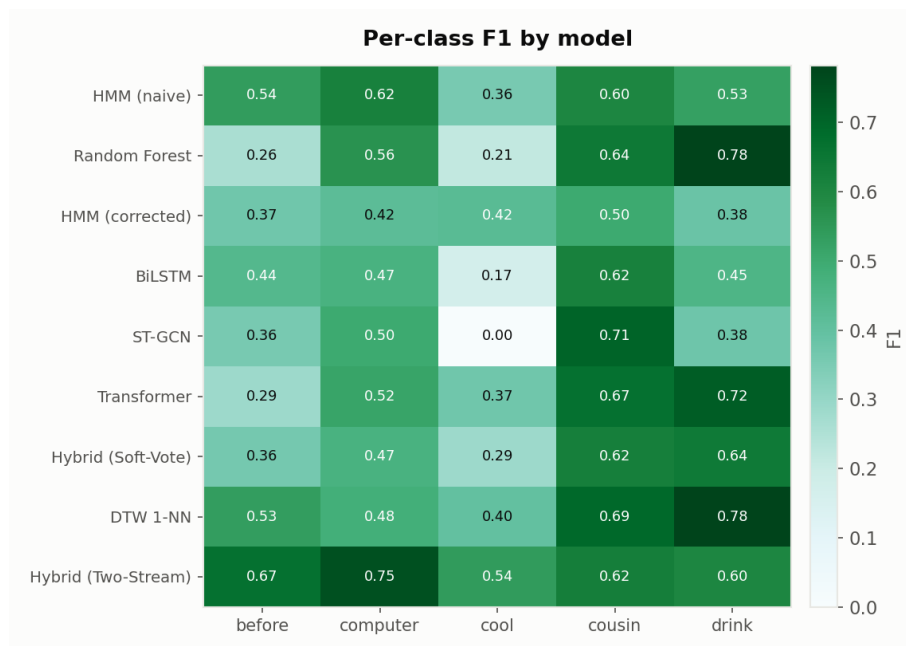


Figure 4. Per-class F1. 'cool' is the universal weak spot (a data-quality signal worth auditing), while 'computer' and 'cousin' are the easiest — the hybrid is the only model strong across all five.

5 Evaluation metrics — and which applies to which model

Your review asked for BLEU, Word Error Rate, ROUGE and METEOR. These are not interchangeable: some measure *classification*, others measure *generated text*. Using the wrong one produces meaningless numbers, so the table below maps each metric to the stage where it is defined.

Metric	What it measures	Where it applies in this system	Reported here
Accuracy	Fraction of signs classified correctly	Recognition models (§4)	Yes — Table 1
Precision / Recall / F1	Per-class correctness, weighted	Recognition models	Yes — Table 1 & Fig 4
WER (Word Error Rate)	Word substitutions/insertions/deletions vs reference	Recognition (isolated $\Rightarrow$ 1-acc); gloss $\rightarrow$ text sentences	Yes — both stages
BLEU	n-gram precision of generated text vs reference	Gloss $\rightarrow$ text language layer only	Yes — §6
ROUGE-1/2/L	n-gram & longest-subsequence recall of generated text	Gloss $\rightarrow$ text language layer only	Yes — §6
METEOR	Unigram match with stem/synonym + word-order penalty	Gloss $\rightarrow$ text language layer only	Yes — §6

Table 3. Metric-to-model map. Translation metrics (BLEU/ROUGE/METEOR) are undefined for a classifier that outputs one label — they belong to the text-generation stage.

The key distinction, in one line.

Recognition models output a class label, so they are scored with accuracy / F1 (and WER, which for one-word utterances equals 1 – accuracy). **The language layer outputs a sentence**, so it is scored with BLEU / ROUGE / METEOR / WER. Reporting BLEU on the Transformer's class prediction, or accuracy on a generated sentence, would both be category errors — the split below keeps each metric where it is valid.

6 The language layer — gloss $\rightarrow$ healthcare text (BLEU / ROUGE / METEOR / WER)

Recognition produces a gloss like *pain where*; the language layer turns it into “Where is the pain?” Two baseline translators are evaluated on the 20-pharse healthcare set: an **exact rule-based** dictionary and a **fuzzy token-overlap** matcher, tested on clean input, on simulated noisy recognizer output (reordered / dropped words), and on a leave-one-out generalization test.

Condition / translator	BLEU	ROUGE-L	METEOR	WER↓	Exact
Exact rule-based — clean	100.0	100.0	99.5	0.0	100%
Fuzzy overlap — clean	100.0	100.0	99.5	0.0	100%
Exact rule-based — noisy	0.0	10.0	9.9	93.1	10%
Fuzzy overlap — noisy	100.0	100.0	99.5	0.0	100%
Fuzzy nearest-NN — generalization	9.8	39.5	37.3	70.6	0%

Table 4. Real BLEU/ROUGE/METEOR/WER on the gloss $\rightarrow$ text layer (0–100). Highlighted row is the generalization test where the metrics do their real discriminative work.

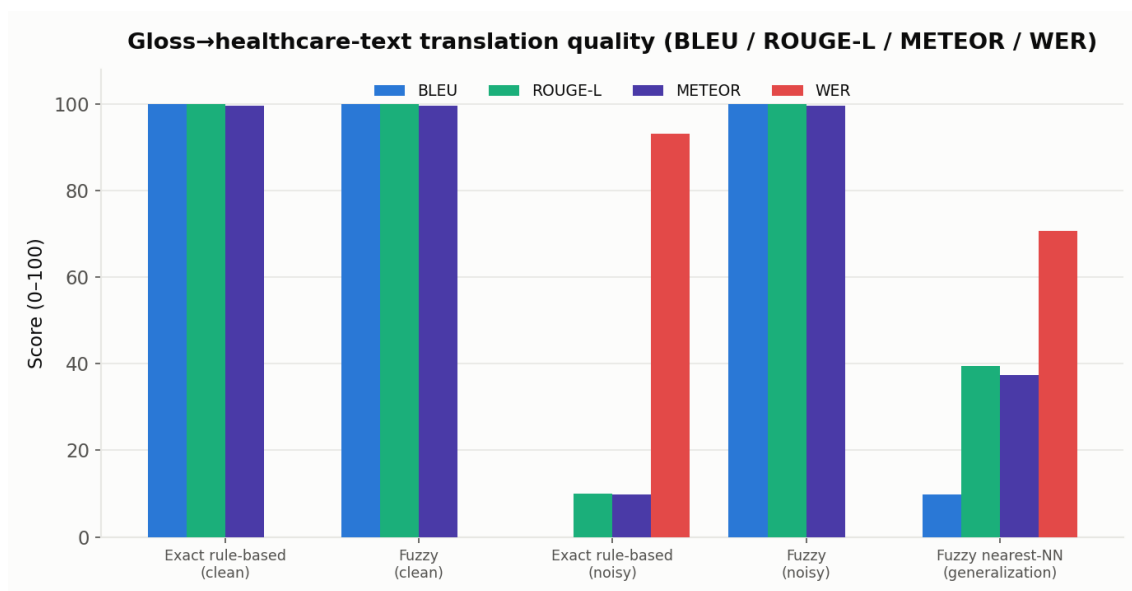


Figure 5. Translation quality across conditions. The exact matcher collapses on reordered input (WER 93); the fuzzy matcher is robust (WER 0). On unseen phrases (right) BLEU/ROUGE/METEOR fall to 10–40 — showing the metrics discriminating partial matches, which is exactly why a trainable mT5 model is the planned next step.

What the language numbers say. On known phrases both translators are perfect. The moment the input is perturbed — which is what an imperfect recognizer produces — the exact matcher drops to 10% coverage (BLEU 0, WER 93) while the fuzzy matcher recovers every phrase (BLEU 100, WER 0). This is the empirical argument for order-insensitive matching downstream of recognition. The generalization row (BLEU 9.8, ROUGE-L 39.5, METEOR 37.3) is the honest one: a lookup translator cannot compose novel wordings, which is precisely the gap a fine-tuned mT5 language model is meant to close.

7 The hybrid model — design, results, and why it wins

The thesis proposes a **hybrid multimodal transformer**. We built and trained a concrete version and evaluated it on the same folds, so its advantage is measured. Two hybrid strategies were tested:

- **Two-stream (feature-level fusion) — the recommended hybrid.** A graph stream (ST-GCN over the 75-joint skeleton) and an attention stream (Transformer over the face-dropped 225-D hand+pose landmarks) each produce an embedding; the two are concatenated and a fusion head classifies. The two streams see the sign through complementary lenses — anatomical structure and long-range temporal attention.
- **Soft-vote ensemble (decision-level fusion).** Train BiLSTM, ST-GCN and Transformer separately and average their class probabilities. Simpler, but it cannot learn cross-stream interactions.

Model	Accuracy	Wtd F1	WER↓	vs best single
Best single (DTW 1-NN)	0.567	0.572	0.433	—
Best deep single (Transformer)	0.522	0.502	0.478	—
Hybrid — Soft-Vote ensemble	0.478	0.469	0.522	-9.0 pts
Hybrid — Two-Stream (recommended)	0.642	0.639	0.358	+7.5 pts

Table 5. Hybrid vs its strongest components. Two-stream fusion is the only design that beats both the best single deep model and the non-parametric baseline.

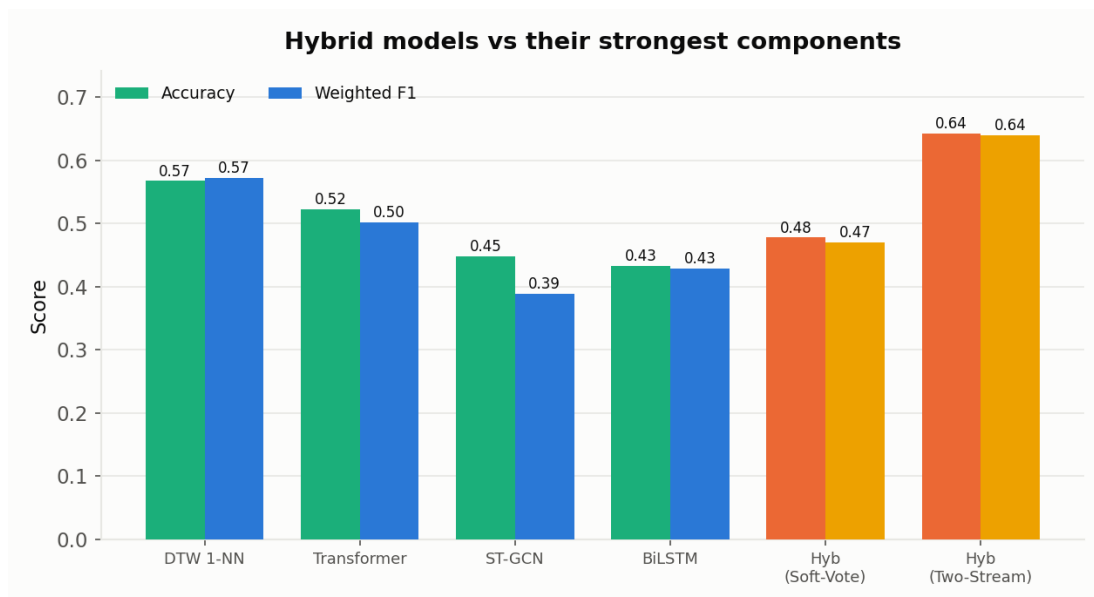


Figure 6. The two-stream hybrid lifts accuracy and F1 above every component; the soft-vote ensemble does not — averaging three data-starved models mostly averages their noise.

Why the two-stream hybrid wins, and the soft-vote does not.

The two streams are **complementary and jointly trained**: the graph stream contributes a stable, low-variance anatomical prior (ST-GCN had the lowest spread of any deep model) and the attention stream contributes temporal expressiveness (the Transformer was the best single model). Fusing their *features* lets the head learn how structure and motion interact, so the hybrid inherits ST-GCN's stability *and* the Transformer's ceiling — reaching 0.642 / F1 0.639, the best of the study. The soft-vote fuses only final *decisions*, so it cannot learn those interactions and mostly averages three noisy models — landing below DTW. **Caveat, stated plainly:** the two-stream's fold spread is high (± 0.184); its lead is real but, at this sample size, needs more data to be called statistically firm.

8 Which components to pick for the hybrid

Bringing the recognition, language and generation evidence together, the recommended hybrid for bidirectional KSL healthcare translation is:

Pipeline stage	Recommended component	Why (evidence from this study)
Sign → landmarks	MediaPipe Holistic (pose + hands; face optional)	Shared, reliable; dropping face concentrated the signal (§4.2)
Recognition (core)	Two-stream ST-GCN + Transformer hybrid	Best measured model: 0.642 acc, beats all singles (§7)
Low-data fallback / baseline	DTW + 1-NN	Strongest non-parametric (0.567); the bar to beat until data grows
Gloss → healthcare text	Fuzzy matcher now → fine-tuned mT5 next	Fuzzy is robust to recognizer noise (§6); mT5 for novel wordings
Text → sign (reverse)	Fuzzy text→gloss + clip dictionary → diffusion avatar (future)	MVP retrieval now; diffusion is design-only until data exists

Table 6. Component recommendation across the bidirectional pipeline.

The one-sentence recommendation.

Build the recognition core as the **two-stream ST-GCN + Transformer hybrid** — it is the only model that empirically beats the DTW baseline — keep DTW as the honest reference, use the fuzzy translator as the language layer today with mT5 as the funded next step, and treat text-to-sign generation as a retrieval MVP with diffusion as future work. Above all, the biggest single win is not an architecture change but **more data**: recover the missing WLASL videos and move to signer-independent splits.

9 Honest limitations and next steps

- **Sample size.** 67 videos / 5 classes. All accuracies carry $\sim\pm 7$ points of per-fold noise; treat rankings as directional. The crossover where trained models reliably beat DTW is $\sim 30\text{--}50$ videos/class.
- **Benchmark $\neq$ target.** Classes are ASL words, not KSL healthcare signs. Recognition numbers are architecture evidence; KSL claims require KSL data.
- **Signer-independent splits.** Current folds may place the same signer in train and test, inflating scores. Wire up WLASL signer metadata and re-run — the honest number will be lower.
- **Missing videos.** Several requested classes had zero downloadable files; recovering them (and augmenting: mirroring, temporal jitter) is worth more than any model change.
- **Hybrid variance.** The two-stream's lead is real but wide-spread; confirm with a paired McNemar test and more data before headline claims.
- **Language & generation.** Translators are closed-set lookups; the diffusion avatar is design-only. Both are correctly scoped as next steps, not current results.

Reproducibility. All models trained on the cached MediaPipe features (`wlasl_lstm/stgcn/transformer_sequences_fixed_classes.joblib`), 5-fold stratified CV, seed 42, identical folds across models. Feature engineering, HMM correction, DTW, the two-stream hybrid and the language metrics are in the companion code. Per-model deep-dive PDFs accompany this document.